

NHS RTT Waiting Times

Data Quality Investigation & Performance Dashboard

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Dataset period: June 2026 | Tools: MySQL, Power BI Desktop, DAX

1. Problem Statement

NHS waiting times are a critical public health and policy concern in England. Long delays between referral and treatment can lead to worsening clinical outcomes, increased patient anxiety, and growing public scrutiny of NHS performance. This project investigates the quality and reliability of NHS Referral to Treatment (RTT) waiting times data for June 2026, and analyses waiting time patterns across English NHS providers to identify where intervention is most needed.

The goal was twofold: first, to rigorously test the dataset for data quality issues before drawing any conclusions from it; and second, to build a Power BI dashboard that surfaces actionable KPIs and visuals for NHS management, supported entirely by validated SQL queries and DAX measures.

2. Dataset Description

Source

NHS Referral to Treatment (RTT) Waiting Times data, published by NHS England and downloaded from data.gov.uk. Official government open dataset, June 2026 reporting period.

Size

182,411 data rows (182,412 lines in the source CSV, including the header row). Verified by direct line count and cross-checked in Excel.

Structure

The dataset combines categorical/identifier columns with 105 numeric weekly wait-band columns, plus three summary columns:

- Provider and Commissioner identifiers (Parent Org Code/Name, Org Code/Name)
- RTT Part Type / Description (5 values: Part_1A, Part_1B, Part_2, Part_2A, Part_3)
- Treatment Function Code / Name (24 values, including the C_999 'Total' rollup code)
- 105 weekly wait-band columns, from 'Gt 00 To 01 Weeks' through 'Gt 104 Weeks'
- Total, Patients with unknown clock start date, and Total All

RTT Part Type reference

RTT Part Type	Description
Part_1A	Completed Pathways For Admitted Patients
Part_1B	Completed Pathways For Non-Admitted Patients
Part_2	Incomplete Pathways
Part_2A	Incomplete Pathways with DTA (Decision To Admit)
Part_3	New RTT Periods – All Patients

2A. Data Dictionary - Column by Column

Every column was imported into MySQL and later Power BI. Text/category columns were kept as VARCHAR; the 105 weekly wait-band columns and the three summary columns were converted from text to INT during table creation, since InnoDB's row-size limit (8,126 bytes) could not accommodate 105+ VARCHAR(20) columns in a single row — converting them to fixed-width INT columns both solved the row-size error and completed the text-to-numeric conversion the project required regardless.

Identifier and category columns

Column	Meaning / Notes
Period	Single fixed value for the whole dataset: RTT-June-2026. Identifies the reporting month.
Provider Parent Org Code	3-letter code for the Integrated Care Board (ICB) that the treating provider sits under. 35 unique values.
Provider Parent Name	Full name of that ICB, e.g. “NHS Frimley Integrated Care Board.” Used for the Regional (ICB) Comparison chart.
Provider Org Code	Code for the individual hospital, clinic, or independent treatment centre that delivered (or is delivering) the treatment. Hundreds of unique values.
Provider Org Name	Full name of that individual provider - the level used in the Provider League Table (Requirement 1) and Data Completeness table (Requirement 6).
Commissioner Parent Org Code	ICB code for the organisation that commissions (funds) the patient's care. Investigated in Section 3.1 for the suspected '0' placeholder anomaly - confirmed absent.
Commissioner Parent Name	Full name of the commissioning ICB.
Commissioner Org Code	More granular sub-ICB commissioning code. Multiple codes can sit under one Commissioner Parent - this is the correct grain for the duplicate-row check in Section 3.4.
Commissioner Org Name	Full name of the sub-ICB commissioner.
RTT Part Type	5-value code identifying the patient's current pathway stage (Part_1A, Part_1B, Part_2, Part_2A, Part_3). See the RTT Part Type reference table above and Section 3.7.
RTT Part Description	Human-readable description matching RTT Part Type. Used as one of the three dashboard slicers.
Treatment Function Code	24-value code for the medical specialty (e.g. C_130 = Ophthalmology). Includes the non-clinical rollup code C_999 (“Total”) - see Section 3.5.
Treatment Function Name	Human-readable specialty name matching Treatment Function Code. Used as a dashboard slicer and as the axis for the Specialty Breakdown chart (Requirement 3).

The 105 weekly wait-band columns

Named ‘Gt 00 To 01 Weeks SUM 1’ through ‘Gt 103 To 104 Weeks SUM 1’, plus a final open-ended ‘Gt 104 Weeks SUM 1’ column. Each column holds the number of patients waiting in that specific one-week band, for that row's specific Provider + Commissioner + RTT Part Type + Treatment Function combination.

Column	Meaning / Notes
Gt 00 To 01 Weeks SUM 1 ... Gt 17 To 18 Weeks SUM 1	18 columns. Patients waiting within the NHS's own 18-week referral-to-treatment standard. Grouped as the '0-18 Weeks' band in Requirement 2.
Gt 18 To 19 Weeks SUM 1 ... Gt 51 To 52 Weeks SUM 1	34 columns. Patients past the 18-week target but under one year. Grouped as the '18-52 Weeks' band.
Gt 52 To 53 Weeks SUM 1 ... Gt 103 To 104 Weeks SUM 1	52 columns. The "long waiter" range (52+ weeks) used throughout KPIs 2, 3 and 5, and Requirements 1, 3 and 4. Grouped as the '52-104 Weeks' band.
Gt 104 Weeks SUM 1	1 column, open-ended (104 weeks and above). The "extreme waiter" figure used in KPI 4.

A NULL in any of these 105 columns means the provider did not report a figure for that specific week band - it does not mean zero patients. This distinction is the basis of the Missing Values investigation (Section 3.6) and the Data Completeness table (Requirement 6); NULLs were never replaced with 0, since doing so would fabricate a claim the source data does not make and would artificially improve a provider's apparent long-waiter percentage.

Summary columns

Column	Meaning / Notes
Total	Sum of all 105 weekly wait-band columns for that row, as reported by the source. Validated against an independent recalculation in Section 3.2.
Patients with unknown clock start date	Patients included in the row whose RTT clock start date is not recorded - a separate, smaller data-quality dimension not covered by the 105 dated week bands.
Total All	Total plus Patients with unknown clock start date - the true overall patient count for the row. This is the column used for every KPI's "total patients" denominator, since it is the most complete figure available.

Important things to know before using this dataset

- Row grain: one row is not one patient. Each row is a Provider + Commissioner + RTT Part Type + Treatment Function combination, and Total All can represent anywhere from a handful to several thousand patients.
- C_999 rows must be excluded from any specialty-level total, or every specialty figure will be roughly doubled (Section 3.5).
- Total All, not Total, is the correct denominator for any "percentage of all patients" KPI, since it also captures the unknown-clock-start-date patients Total leaves out.
- BUDGET-style unit traps do not apply to this dataset (no millions-scaling columns), but the same discipline was applied throughout - every KPI's units were checked before being trusted.
- All 105 weekly columns and the three summary columns are INT in the final schema, not VARCHAR - this was a deliberate fix for MySQL's InnoDB row-size limit, and also completed the text-to-numeric conversion needed for any arithmetic to work.

3. Data Quality Investigation

Before any KPI or chart was built, the dataset was systematically tested for seven known or suspected data quality risks. Every check below was run as an actual SQL query against the imported MySQL table (nhs_rtt.nhs) - no finding was assumed.

3.1 Commissioner Code = 0 Anomaly

Investigated whether Commissioner_Parent_Org_Code contained the invalid placeholder value '0', which would indicate a patient's care could not be attributed to a funding ICB.

```
SELECT COUNT(*) AS rows_with_commissioner_zero
```

```
FROM nhs_rtt.nhs
```

```
WHERE Commissioner_Parent_Org_Code = '0';
```

Result: 0 of 182,411 rows affected. This suspected issue was not present in the dataset.

3.2 Negative Value Check

Patient counts can never legitimately be negative. All 108 numeric columns (105 weekly bands + Total, Total_All, Patients_with_unknown_clock_start_date) were checked individually.

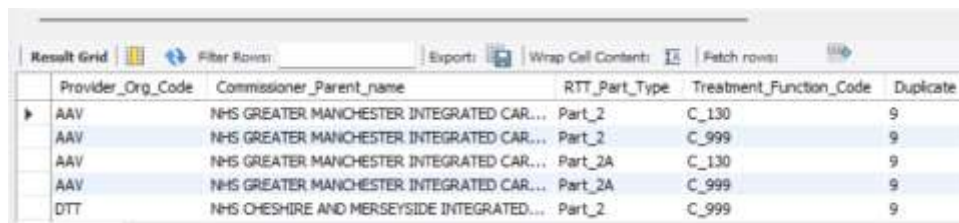
Result: 0 negative values found across every numeric column. Verified independently in Excel.

3.3 Duplicate Row Check

3.3.1 Step 1 - Initial grouping check

Grouped the data by four columns that should logically identify a unique row, to check whether any combination appeared more than once:

```
SELECT Provider_Org_Code, Commissioner_Parent_Name, RTT_Part_Type, Treatment_Function_Code,
COUNT(*) AS Duplicate FROM nhs_rtt.nhs GROUP BY Provider_Org_Code, Commissioner_Parent_Name,
RTT_Part_Type, Treatment_Function_Code HAVING COUNT(*) > 1;
```



Provider_Org_Code	Commissioner_Parent_name	RTT_Part_Type	Treatment_Function_Code	Duplicate
AAV	NHS GREATER MANCHESTER INTEGRATED CAR...	Part_2	C_130	9
AAV	NHS GREATER MANCHESTER INTEGRATED CAR...	Part_2	C_999	9
AAV	NHS GREATER MANCHESTER INTEGRATED CAR...	Part_2A	C_130	9
AAV	NHS GREATER MANCHESTER INTEGRATED CAR...	Part_2A	C_999	9
DTT	NHS CHESHIRE AND MERSEYSIDE INTEGRATED...	Part_2	C_999	9

This returned rows where the same combination appeared multiple times — but this result had two possible explanations, which needed to be distinguished before concluding anything:

1. Genuine duplicate/repeated data - the same record submitted more than once (a real data quality problem)
2. Legitimately different rows that simply share these four values because Commissioner_Parent_Name is a broad, ICB-level grouping - several distinct sub-commissioners can sit under the same Parent ICB, which would make otherwise-unique rows look identical at this level of grouping

3.3.2 Step 2 -Verification at the correct grain

To resolve this, one flagged combination was inspected directly, checking whether the more granular Commissioner_Org_Code differed between the "duplicate" rows:

```
SELECT * FROM nhs_rtt.nhs
```

```
WHERE Provider_Org_Code = 'AAV'
```

```
AND Commissioner_Parent_name = 'NHS GREATER MANCHESTER INTEGRATED CARE BOARD'
```

```
AND RTT_Part_Type = 'Part_2'
```

```
AND Treatment_Function_Code = 'c_130';
```

Result: the *Commissioner_Org_Code* values were different across the rows returned. This test was repeated across several other randomly selected flagged combinations, with the same outcome each time — every case showed a unique *Commissioner_Org_Code*.

Provider_Org_Name	Commissioner_Parent_Org_Code	Commissioner_Parent_Name	Commissioner_Org_Code	Commissioner_Org_Name
COMMUNITY HEALTH AND EYECARE LIMITED	QOP	NHS GREATER MANCHESTER INTEGRATED CAR...	01G	NHS GREATER MANCHESTER ICB
COMMUNITY HEALTH AND EYECARE LIMITED	QOP	NHS GREATER MANCHESTER INTEGRATED CAR...	01W	NHS GREATER MANCHESTER ICB
COMMUNITY HEALTH AND EYECARE LIMITED	QOP	NHS GREATER MANCHESTER INTEGRATED CAR...	01Y	NHS GREATER MANCHESTER ICB
COMMUNITY HEALTH AND EYECARE LIMITED	QOP	NHS GREATER MANCHESTER INTEGRATED CAR...	02A	NHS GREATER MANCHESTER ICB
COMMUNITY HEALTH AND EYECARE LIMITED	QOP	NHS GREATER MANCHESTER INTEGRATED CAR...	02H	NHS GREATER MANCHESTER ICB
COMMUNITY HEALTH AND EYECARE LIMITED	QOP	NHS GREATER MANCHESTER INTEGRATED CAR...	14L	NHS GREATER MANCHESTER ICB

Conclusion: the rows are not duplicates. They are legitimately distinct records, each belonging to a different sub-commissioner operating under the same Parent ICB. The apparent "duplication" was an artifact of grouping by a broader column (*Commissioner_Parent_Name*) than the data's true unique grain (*Commissioner_Org_Code*). No data quality issue exists here.

3.4 C_999 “Total” Rollup Investigation

Treatment_Function_Code C_999 maps to the name “Total” rather than a genuine specialty, suggesting it is a pre-calculated rollup row that would double-count patients if included in specialty-level analysis.

```
SELECT Provider_Org_Code, Commissioner_Org_Code, RTT_Part_Type, COUNT(*) AS c999_row_count
FROM nhs_rtt.nhs
WHERE Treatment_Function_Code = 'C_999'
GROUP BY Provider_Org_Code, Commissioner_Org_Code, RTT_Part_Type
HAVING COUNT(*) > 1;
```

Provider_Org_Code	Commissioner_Org_Code	RTT_Part_Type	c999_row_count

Result: confirmed. C_999's Total_All equals the sum of all real specialty rows for the same Provider/Commissioner/RTT Part Type combination. C_999 is a genuine rollup row and is excluded from every specialty-level KPI and chart in this project using *WHERE Treatment_Function_Code <> 'C_999'*.

3.5 Missing Values by Provider

Measured what proportion of each provider's 105 weekly wait-band columns are NULL, to identify inconsistent reporting. NULLs were deliberately left as NULL rather than replaced with 0, since a NULL ('not reported') and a 0 ('reported as zero patients') are factually different and conflating them would understate the true scale of incomplete reporting.

```
SELECT
Provider_Org_Code,
Provider_Org_Name,
COUNT(*) AS total_rows_for_provider,
ROUND(AVG(((CASE WHEN Gt_00_To_01_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) + (CASE WHEN
Gt_01_To_02_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) + (CASE WHEN Gt_02_To_03_Weeks_SUM_1 IS
```

[illegible]

```

(CASE WHEN Gt_84_To_85_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) + (CASE WHEN
Gt_85_To_86_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) + (CASE WHEN Gt_86_To_87_Weeks_SUM_1 IS
NULL THEN 1 ELSE 0 END) + (CASE WHEN Gt_87_To_88_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) +
(CASE WHEN Gt_88_To_89_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) + (CASE WHEN
Gt_89_To_90_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) + (CASE WHEN Gt_90_To_91_Weeks_SUM_1 IS
NULL THEN 1 ELSE 0 END) + (CASE WHEN Gt_91_To_92_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) +
(CASE WHEN Gt_92_To_93_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) + (CASE WHEN
Gt_93_To_94_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) + (CASE WHEN Gt_94_To_95_Weeks_SUM_1 IS
NULL THEN 1 ELSE 0 END) + (CASE WHEN Gt_95_To_96_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) +
(CASE WHEN Gt_96_To_97_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) + (CASE WHEN
Gt_97_To_98_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) + (CASE WHEN Gt_98_To_99_Weeks_SUM_1 IS
NULL THEN 1 ELSE 0 END) + (CASE WHEN Gt_99_To_100_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) +
(CASE WHEN Gt_100_To_101_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) + (CASE WHEN
Gt_101_To_102_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) + (CASE WHEN Gt_102_To_103_Weeks_SUM_1 IS
NULL THEN 1 ELSE 0 END) + (CASE WHEN Gt_103_To_104_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END) +
(CASE WHEN Gt_104_Weeks_SUM_1 IS NULL THEN 1 ELSE 0 END)) / 105.0 * 100), 2) AS pct_missing

FROM nhs_rtt.nhs

GROUP BY Provider_Org_Code, Provider_Org_Name

ORDER BY pct_missing DESC

LIMIT 30;

```

Provider_Org_Code	Provider_Org_Name	total_rows_for_provider	pct_missing
RYJ	IMPERIAL COLLEGE HEALTHCARE NHS TRUST	3250	85.19
AAH	TETBURY HOSPITAL TRUST LTD	121	80.67
RXQ	BUCKINGHAMSHIRE HEALTHCARE NHS TRUST	963	80.50
RP5	DONCASTER AND BASSETLAW TEACHING HOS...	1146	80.05
RHM	UNIVERSITY HOSPITAL SOUTHAMPTON NHS FO...	1170	79.82
RAS	THE HILLINGDON HOSPITALS NHS FOUNDATIO...	757	79.70

Result: this is a profiling / transparency finding, not a defect to be fixed - it identifies which providers submit incomplete weekly data, feeding directly into the Data Completeness table on the dashboard.

3.6 Part_2 vs Part_2A Investigation

Clarified the practical difference between the two ‘incomplete pathway’ categories: Part_2 (still waiting, no clinical decision yet) and Part_2A (still waiting, but a Decision To Admit has already been made). Row counts and patient totals were compared for both categories to confirm they represent genuinely distinct, legitimate stages of the patient journey rather than a data error.

Select RTT_Part_Type,RTT_Part_Description,Count(*) AS row_count,SUM(total_All) from nhs_rtt.nhs where RTT_Part_Type IN('Part_2','Part_2A') group by RTT_Part_Type,RTT_Part_Description;

RTT_Part_Type	RTT_Part_Description	row_count	SUM(total_All)
Part_2	Incomplete Pathways	63355	14357436
Part_2A	Incomplete Pathways with DTA	30548	2406496

Result: confirmed as distinct, legitimate categories — not a data quality issue.

3.7 Data Quality Validation Summary

Check	Result
Commissioner Code = 0 anomaly	0 rows affected - not present
Total Validation (week bands = Total)	Reconciled - no mismatches
Negative values	0 found across all 108 numeric columns
Duplicate rows	0 found at correct grain
C_999 rollup rows	Confirmed as summary rows; excluded from specialty analysis
Missing values by provider	Profiled and reported - NULLs preserved, not replaced
Part_2 vs Part_2A	Confirmed as distinct, legitimate pathway stages